



NAME: _____

TEACHER: _____

GOSFORD HIGH SCHOOL

2012/2013

EXTENSION 2 MATHEMATICS

HSC ASSESSMENT TASK 1.

Time Allowed: 60 minutes (plus 5 min. reading time)

- Write using black or blue pen.
- Board-approved calculators may be used.
- Section III should be started on a new page and Section IV should be started on a new page.
- All necessary working should be shown in Section II, III and IV.

Papers are to be handed up in 3 bundles. Section I and II in one bundle, Section III in another and Section IV in another.

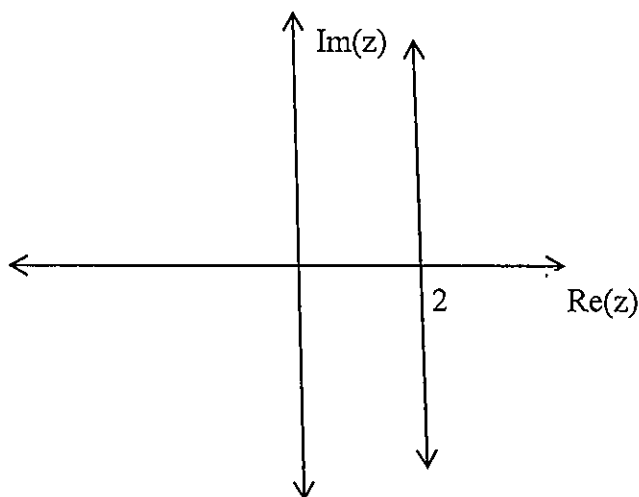
SECTION	QUESTION TYPE	MARKS	RESULT
I	MULTIPLE CHOICE	4	
II	EXTENDED RESPONSE	12	
III	EXTENDED RESPONSE	12	
IV	EXTENDED RESPONSE	12	
	TOTAL	40	

SECTION I. (4 marks) Answer on your own paper by writing down the correct letter.

1. If $z = 3 - i$, what is the value of $\bar{z}$?

- A. $-1 - 3i$ B. $-1 + 3i$ C. $1 - 3i$ D. $1 + 3i$

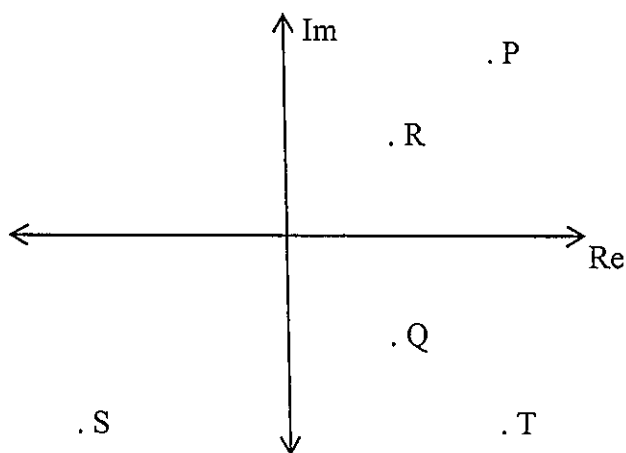
2.



Which of the following is **not** a valid algebraic description of the locus drawn above?

- A. $Re(z) = 2$ B. $|z| = |z - 4|$ C. $Arg(z - 4) + Arg(z) = \pi$ D. $z + \bar{z} = 4$

3. If the point P represents $\omega = 1 + i$, which of the following points best represents $\frac{1}{\omega}$?



- A. Q B. R C. S D. T

4. If $z_1 = 2 \operatorname{cis} \left(\frac{5\pi}{6} \right)$ and $z_2 = 4 \operatorname{cis} \left(\frac{\pi}{6} \right)$ then $\frac{z_1}{z_2} =$

- A. $\frac{1}{2} \operatorname{cis} \left(\frac{2\pi}{3} \right)$ B. $2 \operatorname{cis} \left(-\frac{2\pi}{3} \right)$ C. $\frac{1}{2} \operatorname{cis} \left(\frac{5\pi^2}{36} \right)$ D. $8 \operatorname{cis} (\pi)$

SECTION II. (12 marks)

1. Let $z = 3 - 2i$ and $\omega = 1 - i$. Find in the form $x + iy$, where x and y are real,

(i) $2z + i\omega$ (1)

(ii) $\bar{z}\omega$ (1)

(iii) $\frac{4}{\omega}$ (2)

(iv) $\left|\left(\frac{4}{\omega}\right)\right|$ (1)

2. (i) Express $\frac{1+i\sqrt{3}}{2}$ in modulus-argument form. (2)

(ii) If $z = \frac{1+i\sqrt{3}}{2}$ show that $z^3 = -1$. (1)

(iii) Hence evaluate z^7 , expressing your answer in the form $x + iy$, where x and y are real. (1)

3. (i) Find all the 5th roots of -1 in modulus-argument form. (2)

(ii) Sketch the 5th roots of -1 on an Argand diagram. (1)

SECTION III. (12 marks) Start a new page.

1. (i) Find the complex square roots of $-6i$, expressing your answer in the form $x + iy$, where x and y are real. (2)

- (ii) Solve $z^2 + (1 + i)z + 2i = 0$, expressing the roots z_1 and z_2 in the form $x + iy$, where x and y are real. (2)

- (iii) By letting $\frac{\pi}{12} = \frac{\pi}{4} - \frac{\pi}{6}$ it can be shown that $\tan \frac{\pi}{12} = \frac{\sqrt{3}-1}{\sqrt{3}+1}$. Find a similar expression for $\tan \frac{5\pi}{12}$. (2)

- (iv) Using the results of parts (ii) and (iii), prove that $\arg(z_1) + \arg(z_2) = \frac{\pi}{2}$. (2)

2. If $A = 1 - i$ and $B = 2 + i$ determine the cartesian equation of the locus specified by $|z - A| = |z - B|$ and draw a neat sketch on an Argand diagram to show it. (4)

SECTION IV. (12 marks) Start a new page.

1. (i) On an Argand diagram sketch the locus of $|z| = 2$ and $|z + 2| = 2$. (1)

(ii) Hence or otherwise, find in the form $x + iy$, where x and y are real, all complex numbers simultaneously satisfying $|z| = 2$ and $|z + 2| = 2$. (2)

2. Let $z = \cos\theta + i \sin\theta$

(i) Show that $z^n + z^{-n} = 2 \cos n\theta$. (1)

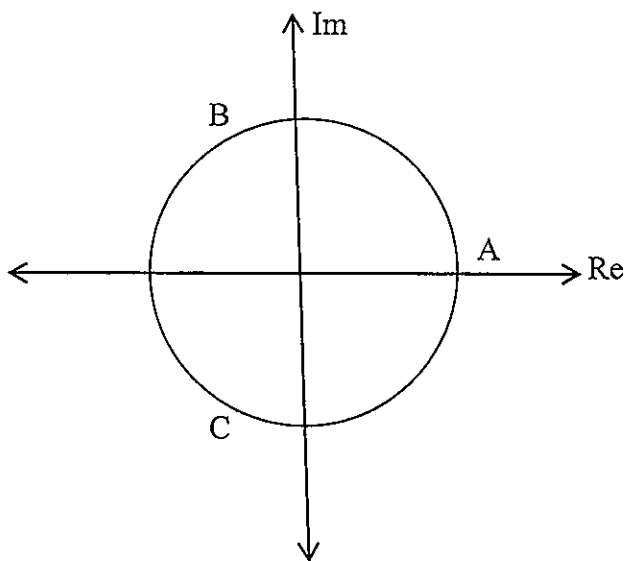
(ii) Hence, or otherwise, show that $\sin^3\theta = \frac{3\sin\theta}{4} - \frac{\sin 3\theta}{4}$. (3)

3. The roots of $z^3 - 1 = 0$ are $1, \omega$ and ω^2 where ω is one of the complex roots.

(i) Find the value of $1 + \omega + \omega^2$. (1)

(ii) Show that $z^2 + z + 1 = (z - \omega)(z - \omega^2)$. (2)

(iii) The Argand diagram shows the points A, B, C on the unit circle which correspond to the roots $1, \omega$ and ω^2 respectively.



Copy this diagram and show the vector $(1 - \omega)$ on your diagram clearly indicating its direction. (1)

(iv) Hence, or otherwise, find the product of the lengths of the chords BA and AC . (1)

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Ex 2

Assess 1/Assess 1

SOLUTIONS

SECTION I

1. $iz = i(3-i)$

$= 3i - i^2$

$= 1 + 3i$

$\therefore \bar{iz} = 1 - 3i$

Hence C

2. If $\operatorname{Re}(z) = 2$

$x = 2 \quad \checkmark$

If $|z| = |z - 4|$

$x = 2 \quad \checkmark$

If $z + \bar{z} = 4$

$x + iy + x - iy = 4$

$2x = 4$

$x = 2 \quad \checkmark$

Hence C

3. If $w = 1 + i$

$\frac{1}{w} = \frac{1}{1+i} \times \frac{1-i}{1-i}$

$= \frac{1-i}{2}$

$= \frac{1}{2} - \frac{1}{2}i$

Hence A

4. $\frac{z_1}{z_2} = \frac{2}{4} \operatorname{cis}\left(\frac{5\pi}{6} - \frac{\pi}{6}\right)$

$= \frac{1}{2} \operatorname{cis} \frac{4\pi}{6}$

$= \frac{1}{2} \operatorname{cis} \frac{2\pi}{3}$

Hence ASECTION II

1. (i) $2z + iw$

$= 2(3-2i) + i(1-i)$

$= 6 - 4i + i - i^2$

$= 7 - 3i$

(1)

(ii) $\bar{z}w$

$= (3+2i)(1-i)$

$= 3 - 3i + 2i - 2i^2$

$= 5 - i$

(1)

(iii) $\frac{4}{w}$

$= \frac{4}{1-i} \times \frac{1+i}{1+i}$

$= \frac{4(1+i)}{1-i^2}$

$= \frac{4(1+i)}{2}$

$= 2 + 2i$

(2)

(iv) $\left| \left(\frac{4}{w} \right) \right|$

$= \sqrt{(2)^2 + (-2)^2} \quad \text{since } \left(\frac{4}{w} \right) = 2 - 2i$

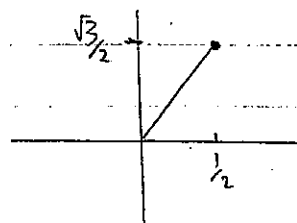
$= \sqrt{8}$

$= 2\sqrt{2}$

(1)

2(i) $\frac{1+i\sqrt{3}}{2}$

$= \frac{1}{2} + \frac{\sqrt{3}}{2}i$



$\therefore r = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}$

$= 1$

$$\theta = \tan^{-1} \left(\frac{\sqrt{3}/2}{1/2} \right)$$

$$= \tan^{-1}(\sqrt{3})$$

$$= \pi/3$$

$$\therefore \frac{1+i\sqrt{3}}{2} = \operatorname{cis} \pi/3 \quad (2)$$

$$(ii) \text{ If } z = \operatorname{cis} \pi/3$$

$$z^3 = \operatorname{cis} 3(\pi/3)$$

$$z^3 = \operatorname{cis} \pi$$

$$= \cos \pi + i \sin \pi$$

$$= -1 + i \cdot 0$$

$$= -1$$

(1)

$$(iii) z^7 = z^6 \cdot z$$

$$= (z^3)^2 \cdot z$$

$$= (-1)^2 \cdot z$$

$$= z$$

$$= \frac{1+i\sqrt{3}}{2} \text{ or } \frac{1}{2} + \frac{\sqrt{3}}{2}i$$

(1)

$$3.6) \text{ Let } z = \cos \theta + i \sin \theta$$

$$\text{If } z^5 = -1$$

$$\cos 5\theta + i \sin 5\theta = -1 + 0i$$

$$\therefore \cos 5\theta = -1$$

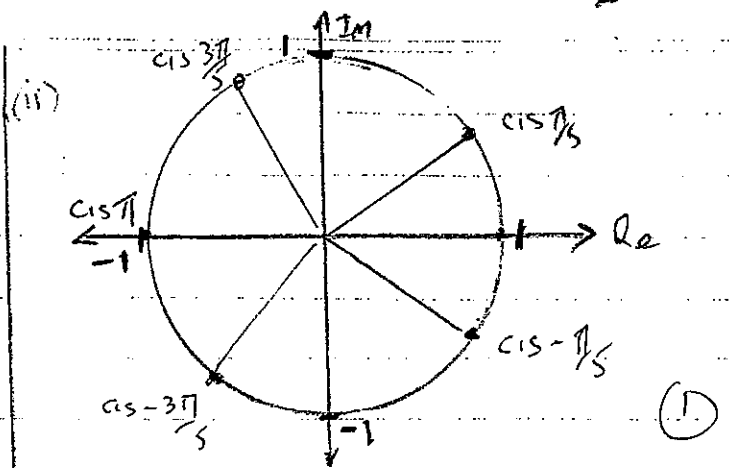
$$5\theta = \pi, 3\pi, 5\pi, 7\pi, 9\pi$$

$$\theta = \frac{\pi}{5}, \frac{3\pi}{5}, \pi, \frac{7\pi}{5}, \frac{9\pi}{5}$$

$$\text{i.e. } \theta = \pm \frac{\pi}{5}, \pm \frac{3\pi}{5}, \pi \quad (2)$$

$$\text{since } \frac{7\pi}{5} = -\frac{3\pi}{5} \text{ and } \frac{9\pi}{5} = -\frac{\pi}{5}$$

$$\text{The roots are } \operatorname{cis} \frac{\pi}{5}, \operatorname{cis} \frac{3\pi}{5}, \operatorname{cis} \pi, \operatorname{cis} -\frac{\pi}{5}, \operatorname{cis} -\frac{3\pi}{5}$$



SECTION II

$$1(i) \text{ Let } \sqrt{-6i} = x+iy$$

$$0-6i = (x+iy)^2$$

$$0-6i = x^2 - y^2 + 2xyi$$

$$x^2 - y^2 = 0, \quad 2xy = -6$$

$$y = -\frac{3}{x}$$

$$\therefore x^2 - \left(-\frac{3}{x}\right)^2 = 0$$

$$x^2 - \frac{9}{x^2} = 0$$

$$x^4 - 9 = 0$$

$$(x^2 - 3)(x^2 + 3) = 0$$

$$x = \pm \sqrt{3}$$

$$y = \mp \frac{3}{\sqrt{3}}$$

$$y = \mp \sqrt{3}$$

(2)

$$\therefore \sqrt{-6i} = \sqrt{3} - i\sqrt{3} \text{ or } -\sqrt{3} + i\sqrt{3}$$

$$(ii) \text{ If } z^2 + (1+i)z + 2i = 0$$

$$z = \frac{-(1+i) \pm \sqrt{(1+i)^2 - 8i}}{2}$$

$$z = \frac{-(1+i) \pm \sqrt{1+2i+i^2-8i}}{2}$$

2

$$= \frac{-(1+i) \pm \sqrt{-6i}}{2}$$

$$= \frac{-(1+i) \pm (\sqrt{3} - i\sqrt{3})}{2}$$

$$= \frac{(\sqrt{3}-1) - (\sqrt{3}+1)i}{2} \quad (2)$$

$$\text{or } \frac{-(\sqrt{3}+1) + (\sqrt{3}-1)i}{2}$$

$$(iii) \quad \frac{5\pi}{12} = \frac{\pi}{6} + \frac{\pi}{4}$$

$$\therefore \tan \frac{5\pi}{12} = \tan \left(\frac{\pi}{6} + \frac{\pi}{4} \right)$$

$$= \frac{\tan \frac{\pi}{6} + \tan \frac{\pi}{4}}{1 - \tan \frac{\pi}{6} \tan \frac{\pi}{4}}$$

$$= \frac{\frac{1}{\sqrt{3}} + 1}{1 - \frac{1}{\sqrt{3}}}$$

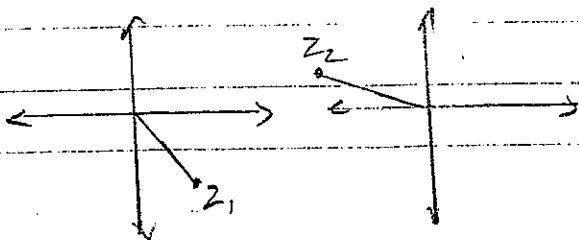
$$= \frac{\sqrt{3}+1}{\sqrt{3}-1}$$

$$= \frac{\sqrt{3}+1}{\sqrt{3}-1}$$

$$= \frac{\sqrt{3}+1}{\sqrt{3}-1} \quad (2)$$

$$(iv) \text{ Let } z_1 = \frac{(\sqrt{3}-1) - (\sqrt{3}+1)i}{2}$$

$$z_2 = \frac{-(\sqrt{3}+1) + (\sqrt{3}-1)i}{2}$$



$$\arg z_1 = \tan^{-1} \left[\frac{-(\sqrt{3}+1)/2}{(\sqrt{3}-1)/2} \right]$$

$$= -\tan^{-1} \left[\frac{\sqrt{3}+1}{\sqrt{3}-1} \right]$$

$$= -\frac{5\pi}{12}$$

$$\arg(z_2) = \pi - \tan^{-1} \left[\frac{(\sqrt{3}-1)/2}{(\sqrt{3}+1)/2} \right]$$

$$= \pi - \frac{\pi}{12}$$

$$= \frac{11\pi}{12}$$

$$\therefore \arg(z_1) + \arg(z_2) = -\frac{5\pi}{12} + \frac{11\pi}{12} = \frac{\pi}{2}$$

$$2. \text{ Let } z = x+iy$$

$$z-A = x+iy - (1-i) = x-1 + i(y+1)$$

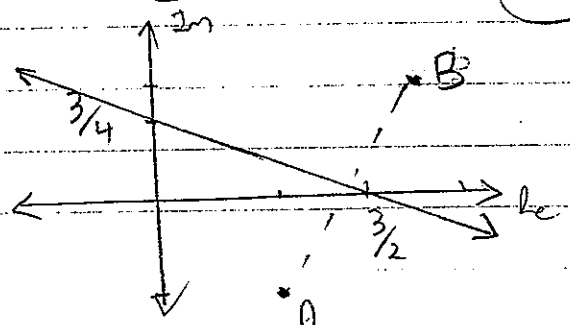
$$z-B = x+iy - (2+i) = x-2 + i(y-1)$$

$$\text{If } |z-A| = |z-B|$$

$$\sqrt{(x-1)^2 + (y+1)^2} = \sqrt{(x-2)^2 + (y-1)^2}$$

$$\therefore x^2 - 2x + 1 + y^2 + 2y + 1 = x^2 - 4x + 4 + y^2 - 2y + 1$$

$$\text{i.e. } 2x + 4y - 3 = 0 \quad (3)$$

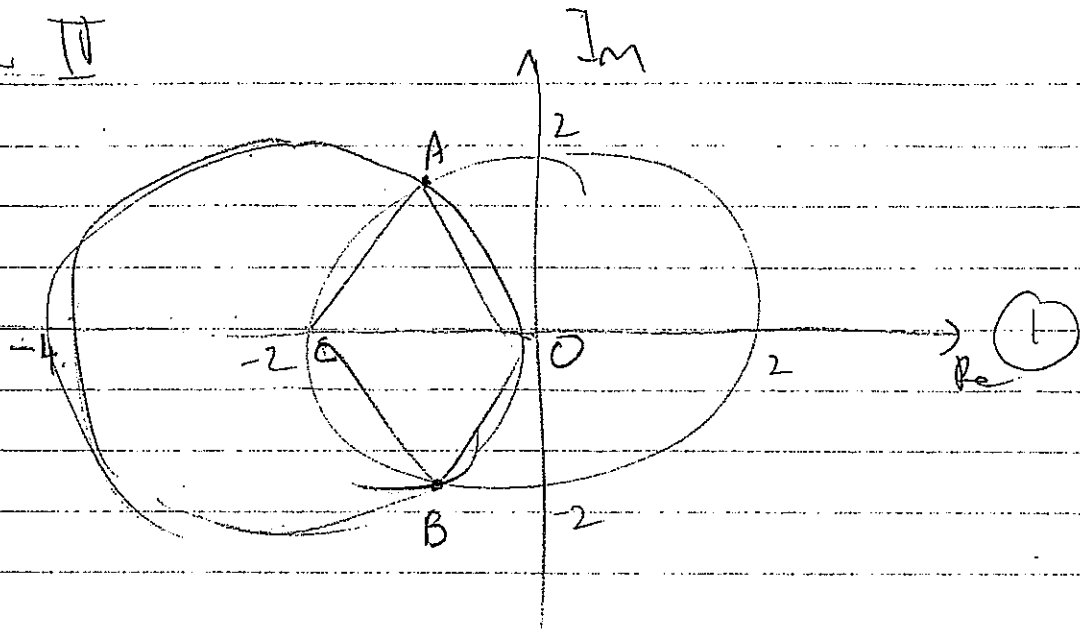


N.B. The locus is the perpendicular bisector of AB

Solution II

1/

(ii)

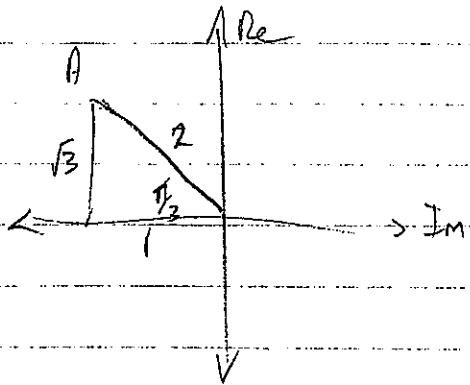


- (ii) Let the points of intersection be A & B
Let the center of the circles be O & C

$\triangle AOC$ & $\triangle BOC$ are equilateral since $OA = OC = OB = OC = BC = 2$ units

$\therefore A$ is the point $\cos \frac{2\pi}{3}$

and B is the point $\cos -\frac{2\pi}{3}$



$\therefore A$ is $-1 + \sqrt{3}i$

by symmetry
 B is $-1 - \sqrt{3}i$

(2)

2/ (i) If $z = \cos \theta + i \sin \theta$

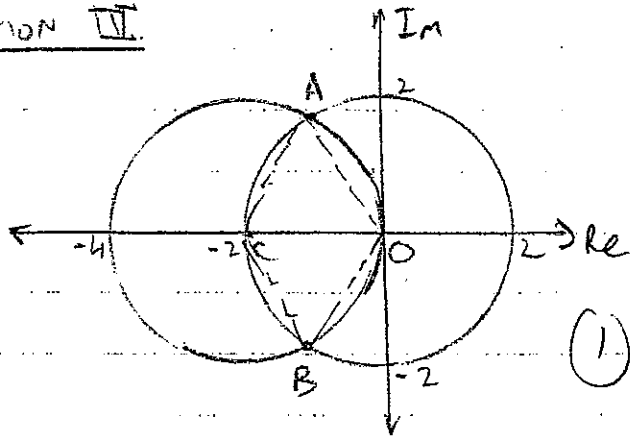
$$z^n = \cos(n\theta) + i \sin(n\theta)$$

$$\begin{aligned} z^{-n} &= \cos(-n\theta) + i \sin(-n\theta) \\ &= \cos(n\theta) - i \sin(n\theta) \end{aligned}$$

$$\therefore z^n + z^{-n} = 2 \cos n\theta$$

SECTION III

1 (i)



(ii)

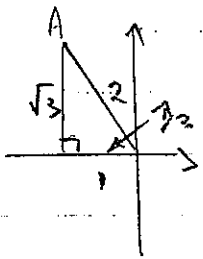
Let the points of intersection be A & B and the centres be O & C

$$OA = OB = OC = CA = CB = 2 \text{ units}$$

$\therefore \Delta AOC$ is equilateral

$\therefore A$ is the point $\text{cis } \frac{2\pi}{3}$

$\therefore B$ is the point $\text{cis } -\frac{2\pi}{3}$



$$A \text{ is } -1 + \sqrt{3}i$$

& by symmetry

$$B \text{ is } -1 - \sqrt{3}i$$

$$2. (i) z^n = \cos(n\theta) + i \sin(n\theta)$$

$$z^{-n} = \cos(-n\theta) + i \sin(-n\theta)$$

$$= \cos(n\theta) - i \sin(n\theta)$$

$$\therefore z^n + z^{-n} = 2 \cos(n\theta) \quad (1)$$

$$(ii) \text{ If } z = \cos \theta + i \sin \theta$$

$$z^n - z^{-n} = 2i \sin(n\theta)$$

$$\therefore z - \frac{1}{z} = 2i \sin \theta$$

$$\text{Now } \left(z - \frac{1}{z}\right)^3 = (2i \sin \theta)^3$$

$$= -8i \sin^3 \theta$$

$$\therefore -8i \sin^3 \theta = z^3 - 3z^2 \cdot \frac{1}{z} + 3z \cdot \frac{1}{z^2} - \frac{1}{z^3}$$

$$-8i \sin^3 \theta = \left(z^3 - \frac{1}{z^3}\right) - 3\left(z - \frac{1}{z}\right)$$

$$-8i \sin^3 \theta = 2i \sin 3\theta - 3(2i \sin \theta)$$

$$-8i \sin^3 \theta = 2i \sin 3\theta - 6i \sin \theta$$

$$\therefore \sin^3 \theta = \frac{2i \sin 3\theta - 6i \sin \theta}{-8i}$$

$$= -\frac{\sin 3\theta}{4} + \frac{3 \sin \theta}{4}$$

$$= \frac{3 \sin \theta}{4} - \frac{\sin 3\theta}{4} \quad (3)$$

$$3. (i) \text{ If } z^3 - 1 = 0$$

$$z^3 + 0z^2 + 0z - 1 = 0$$

The sum of the roots is $-\frac{b}{a}$

$$\therefore 1 + \omega + \omega^2 = 0 \quad (1)$$

$$(ii) (z - \omega)(z - \omega^2)$$

$$= z^2 - z\omega^2 - z\omega + \omega^3$$

$$= z^2 - z(\omega^2 + \omega) + \omega^3$$

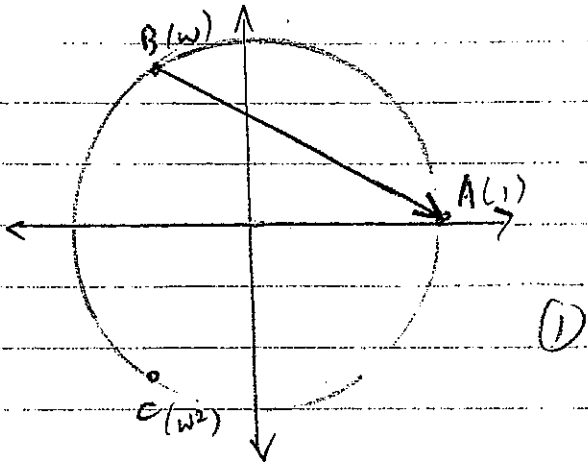
$$= z^2 - z(-1) + 1$$

since $\omega^2 + \omega = -1$ & if ω is a root of $z^3 - 1 = 0$, $\omega^3 = 1$

$$\therefore (z - \omega)(z - \omega^2) = z^2 + z + 1$$

(2)

i)



$$\begin{aligned}
 \text{ii) } \overrightarrow{BA} \cdot \overrightarrow{CA} &= (1-w)(1-w^2) \\
 &= 1 - w^2 - w + w^3 \\
 &= 1 - (w^2 + w) + w^3 \\
 &= 1 - (-1) + 1 \\
 &= 3.
 \end{aligned}$$

①